Sprint 3

In this Sprint, I continued refining the code as we finally got more horses (and eventually all by the end of the sprint), so we were able to host a play test and get some valuable feedback and scope out bugs. The challenge I want to outline the STAR method is how I went about fixing the gyro inputs to get an accurate reading with both rocking and turning going through one gyroscope sensor. I will also outline why we needed to stick with one gyroscope. Another roadblock that didn't cost much time, but was a serious situation to take seriously, was one of our group members not being available after making a plan, causing the pickup of all the horses to be delayed by a day. We were able to work around this, but there was a lesson to be learned for the individual involved, which is important to correct as it could cause more severe consequences in the future.

Situation: We only had enough gyro sensors so that each horse controller got one. During the first playtest, we got feedback that the rocking inputs were causing the turning outputs to be affected too much. Due to budget constraints and space and design, I didn't want to add another gyro (which we initially had) because it would have been too much to figure out how to make it fit and still look clean.

Task: How can I use software/code to make it so the inputs remain consistent and accurate based on whether you are rocking or turning the head?

Action: I coded some cases based on what was observed during playtest, like subtracting the offsets of turning from rocking, so that if you were doing one, it tried not to include the data from the other. I also observed that the data got stuck, so it would never reach the max turn values; if it was stuck, I would ease it to those max values

after a short delay. I also added a deadzone so that it zeroed out if it was close enough to the center point.

Result: Primary testing showed that these changes work, but I need to test it in-game to get accurate results; based on the results, I can tune variables like the delay to ease the values and the deadzone.